

BASIC NUMERICALS FOR IOT - PART 1

Categorization

- Data rate calculations.
- Battery life calculations.
- Latency estimation
- Data Storage requirement
- Range calculation.
- Bandwidth utilization.
- Power Consumption
- Sampling rate calculation.
- Packet loss rate calculation
- Data transmission in Mesh Network.

NUMERICAL - 1

A sensor in an IoT network generates 200 bytes of data in 5 seconds. What is the data rate of this sensor in bits per second (bps)?

Solution : $\text{Data rate} = \frac{\text{Data size (in bits)}}{\text{Time (in second)}}$

$$\begin{aligned}\text{Data size} &= 200 \times 8 \\ &= 1600 \text{ bits}\end{aligned}$$

$$\therefore \text{Data rate} = \frac{1600}{5} = 320 \text{ bps}$$

NUMERICAL 2:

An IoT device consumes 10 mA of current when active and 0.5 mA in sleep mode. If the device is active for 2 hours and sleeps for 22 hrs each day, and the battery capacity is 1000 mAh , what is the battery life in days?

Solution

$$\text{Total Consumption/day} = \text{let } x$$

$$\begin{aligned}\therefore x &= 10 \times 2 + 0.5 \times 22 \\ &= 20 + 11 \\ &= 31\text{ mAh/day}\end{aligned}$$

$$\text{Battery life} = \frac{\text{Battery Capacity}}{x}$$

$$= \frac{1000}{31} \approx 32.26\text{ days.}$$

NUMERICAL 3: An IoT sensor node sends data packets of size 512 bytes to a central server over a network with a bandwidth of 1 Mbps.

Calculate the transmission latency for a single packet.

Solution:

$$\begin{aligned}\text{Packet size} &= 512 \times 8 \text{ bits} \\ &= 4096 \text{ bits}\end{aligned}$$

$$\begin{aligned}\text{Given, bandwidth} &= 1 \text{ Mbps} \\ &= 1000000 \text{ bps}\end{aligned}$$

$$\begin{aligned}\text{Transmission Latency} &= \frac{\text{Packet size}}{\text{Bandwidth}} \\ &= \frac{4096 \text{ bits}}{1000000 \text{ bits/second}} \\ &\approx 4.1 \text{ ms}\end{aligned}$$

NUMERICAL 4: If a temperature sensor

sends data every 10 seconds, and each data packet is 100 bytes, how much data will be stored in 1 day?

Solution:
$$\text{Packets per day} = \frac{24 \times 60 \times 60}{10}$$
$$= 8640$$

$$\therefore \text{Data per day} = 8640 \times 100 \text{ bytes}$$
$$= 864 \text{ KB}$$

NUMERICAL 5: An IoT device operates

at a frequency of 2.4 GHz with a transmission power of 10 dBm. If the signal loss is 100 dB, what is the received signal strength at the receiving end?

Solution:
$$\text{Received signal strength (RSSI)}$$
$$= \text{Transmission Power} - \text{Signal loss}$$
$$= 10 - 100$$
$$= -90 \text{ dBm}$$

NUMERICAL 6: An IoT device transmits

data at 5 Mbps, while the network bandwidth is 20 Mbps. What is the percentage of bandwidth utilization?

Solution:

$$\text{Utilization} = \frac{\text{Device data rate}}{\text{Network BW}} \times 100$$

$$= \frac{5}{20} \times 100 = 25\%$$

NUMERICAL 7: An IoT sensor has an average power consumption of 1.5 mW. If it operates for 12 hrs a day, How much energy does it consume in a month (30 days)?

Solution:

$$\begin{aligned} \text{Energy consumption/day} &= 1.5 \times 12 \\ &= 18 \text{ mWh} \end{aligned}$$

$$\begin{aligned} \text{Energy consumption/month} &= 18 \times 30 \\ &= 540 \text{ mWh} \end{aligned}$$

NUMERICAL 8: A vibration sensor requires a sampling rate of 2000 Hz to capture the data accurately. How many samples will the sensor generate in 1 minute?

Solution:

$$\begin{aligned}\text{Samples per minute} &= 2000 \times 60 \\ &= 120000\end{aligned}$$

NUMERICAL 9: An IoT device sends 1000 packets, but only 920 packets are received successfully. What is the packet loss rate?

Solution:

$$\begin{aligned}\text{Packet loss rate} &= \frac{\text{Sent packets} - \text{Rec. packets}}{\text{Sent packets}} \times 100 \\ &= \frac{1000 - 920}{1000} \times 100 \\ &= 8\%\end{aligned}$$

NUMERICAL 10: In a mesh network of

5 IoT nodes, each node sends 50 KB of data per hour to a central node. Calculate the total data transmitted to the central node in a day?

Solution: Data per node per day

$$\begin{aligned} &= 50 \times 24 \\ &= 1200 \text{ KB} \end{aligned}$$

Total data for 5 nodes

$$\begin{aligned} &= 1200 \times 5 \\ &= 6000 \text{ KB} \\ &= 6 \text{ MB} \end{aligned}$$